

FedShield: Federated Learning-based Privacy-Preserving Intrusion Detection in IoT Networks

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Abstract - Due to the fast growth of the IoT networks, its susceptibility to cyberattacks has grown, and concerns about data privacy and communication overhead of centralized Intrusion Detection Systems (IDS) have been raised. Federated Learning (FL) can also be used for collaborative model training, while ensuring the privacy of the raw data, which is suitable for IoT security. In practice however, the IDS frameworks based on FL can experience performance degradation, given non-Independent and Identically Distributed (non-IID) data, and are not interpretable. In this paper, Federated Intrusion Detection framework FedShield is introduced that combines the Federated Proximal Optimization (FedProx) with SHapley Additive exPlanations (SHAP) to provide strong and explainable threat detection. The proposed framework is tested on the CIC-IoT-2023 dataset, which has 34 classes of network traffic. Five Federated Clients are used to simulate non-IID conditions with a Dirichlet distribution ($\alpha = 0.5$). After 35 communication rounds, experimental results show that FedShield is able to achieve 91.81% accuracy, 0.9089 weighted F1-score, and 0.9961 macro-AUC-ROC. Moreover, SHAP can give explanations about the predictions at the feature level, leading to increased transparency and trustworthiness. The findings show that FedShield can successfully trade-off between preserving privacy, detection performance, and model interpretability for IoT intrusion detection.

Keywords: Federated Learning, Intrusion Detection, IoT Security, FedProx, Explainable AI, SHAP, Privacy-Preserving Learning.

I. INTRODUCTION

The number of Internet of Things (IoT) devices is projected to reach an estimated 30 billion within the next ten years, making it a fairly straightforward task to easily surpass this number [1]. From the consumer smart-home hub to critical sensors in industry, it's a decentralized architecture that is the foundation of digital infrastructure. Unfortunately, the vast majority of these edge devices are designed with severe computation and battery constraints and robust cryptographic defenses on the edge cannot be afforded in most cases [2]. These unprotected nodes are often used to create large botnets which can then be used to conduct a huge Distributed Denial of Service (DDoS) attack or small-scale data exfiltration. With these challenges in mind, the cyber security industry has always relied on centralized machine learning pipelines [3]. For normal deployments, edge gateways report a continuous stream of raw network packet flows and telemetry logs generated in the region to a cloud server located at another site for deep learning analysis. The

centralized detection approach is certainly accurate when it comes to a laboratory setting, but is essentially not appropriate for the scale of IoT today. Delivering terabytes of raw packet data over the network introduces choke-points that slow down the network and provide latency that makes it almost impossible to provide real-time attack mitigation [4]. Moreover – and perhaps more importantly – centralized data pooling is a huge privacy violation. Even metadata that are stripped of the data can be reverse-engineered to expose proprietary operational routines or even users' personal habits, which can be in direct conflict with international privacy laws, such as GDPR [5].

Federated Learning (FL) radically changes this situation [6]. With FL, it is not the data that is transmitted to the model; it is the model that is transmitted to the data. ID algorithm trainer uses localized network traffic for each individual IoT device to train ID algorithm locally. The devices don't share the packets, but rather just the new mathematical weights and biases with a central aggregation server.

FL is wonderfully able to avoid having to deal with the predicament of privacy; however, it is prone to fragility in the face of the chaos of real-world IoT networks. Traffic at the edge devices is not even. An environmental sensor may not receive a rogue packet at all, while a public-facing router may receive such a high volume of packets in a TCP SYN flood that it is unable to keep up. This leads to very highly correlated, and not Independent and Identically Distributed (not IID) data [7]. If models are trained on such vastly different data, and then aggregated using standard aggregation methods like Federated Averaging (FedAvg), the accuracy of the federated model quickly decreases, which is called client drift [8].

Moreover, with the adoption of deep neural networks, an acute “trust gap” has been created in security operations. Intrinsically, deep learning models are a black box. When an automated system identifies a device plugged into the network for quarantine, human security analysts cannot simply take the machine's word for it; they need a logical explanation and justification for disrupting any network services as they would not want to be responsible for disrupting network services intentionally [9]. In this research we propose FedShield, an end-to-end and privacy-preserving intrusion detection architecture, to directly address the non-IID optimization challenge and the human interpretability

gap in an end-to-end manner. This work makes the following contributions:

- We design a strong federated system using FedProx algorithm to effectively counteract the client drift caused by extremely imbalanced and realistic edge settings.
- We make an exhaustive evaluation and classify successfully 34 classes of modern threats considering the contemporary CIC-IoT-2023 dataset [10] without centralization of a single raw data packet.
- Decentralized pipeline with seamless integration of SHapley Additive exPlanations (SHAP) to fill the gap between mathematical optimization and human-centric Security Operations Center (SOC) processes and offer exact feature attributions for each flagged attack.

The rest of the paper goes as follows. A survey of relevant research on federated learning, IoT intrusion detection and explainable AI is presented in Section II. The proposed FedShield framework and methodology is presented in Section III and the experimental setup and performance evaluation is discussed in Section IV. Finally, the paper is ended with some future research directions in Section V.

II. RELATED WORK

Network security research is a very active area, with the move from monolithic cloud-based to decentralized edge-native Intrusion Detection Systems (IDS).

A. Centralized to Decentralized Threat Detection Early

Experiments in foundational studies clearly showed that deep learning outperforms the traditional static signature filtering. Diro and Chilamkurti [4] demonstrated that distributed neural architectures also could detect zero-day anomalies much more effectively than previous rule-based systems. Following this trend, Ferrag et al. [3] made a thorough benchmarking of deep learning models applied in cybersecurity, stating their accuracy, while reporting drawbacks of data centralisation such as latency and privacy.

However, a promising way to break out of the centralization trap was the introduction of Federated Learning by McMahan et al. [6]. In the IoT field, Mothukuri et al. [11] successfully deployed an FL architecture capable of detecting network anomalies from smart devices with high baseline accuracy, and still allowing to preserve privacy of nodes.

In a similar manner, Campos et al. [12] demonstrated the feasibility of FL with regard to N-BaIoT dataset. A huge part of the early FL-IDS literature however, was based on an unrealistic and critical assumption that the training data is IID across all participating clients.

B. Addressing the Non-IID Reality

As the field advanced, researchers realized that the IID assumption is a showstopper for deploying in the edge. Zhao et al. [7] took the first step towards formally documenting the mathematical degradation of neural ensembles when exposed to heterogeneous distribution of local data. This degradation also occurs in IoT malware detection scenarios, as observed by Rey et al. [13]: In the traditional FedAvg, entire minority threat classes are often lost. To mitigate this, Li et al. [8] proposed FedProx, an enhanced optimization algorithm that is a two-step process of local updates followed by a proximal regularization term to link to the global consensus. Although FedProx has been widely adopted in the field of generalized federated learning [14], it is still challenging to measure its effectiveness on a large-scale modern cybersecurity dataset with a large number of classes (34) and under high levels of data skew (Dirichlet distribution).

C. The Imperative of Explainable AI (XAI)

Last but not least, post-hoc explainability is still not well developed in relation to decentralized security networks. Although SHAP [9] has won many hearts in well-regulated industries such as centralized finance and healthcare, the use of SHAP to achieve a high level of explainability in a distributed IoT framework is technically challenging. Aledhari et al. [15] pointed out that edge autonomous security mitigation will never be totally trusted by enterprise administrators if it is not interpretable. FedShield is exactly at this intersection, combining the notions of privacy preservation, heterogeneous stability, and actionable human trust into a single unified pipeline.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

The FedShield infrastructure is a standard star-topology, and is based on the idea of making collaborative machine learning with resource-constrained edge devices through a central aggregation server to guarantee absolute data localization as illustrated in Figure 1.

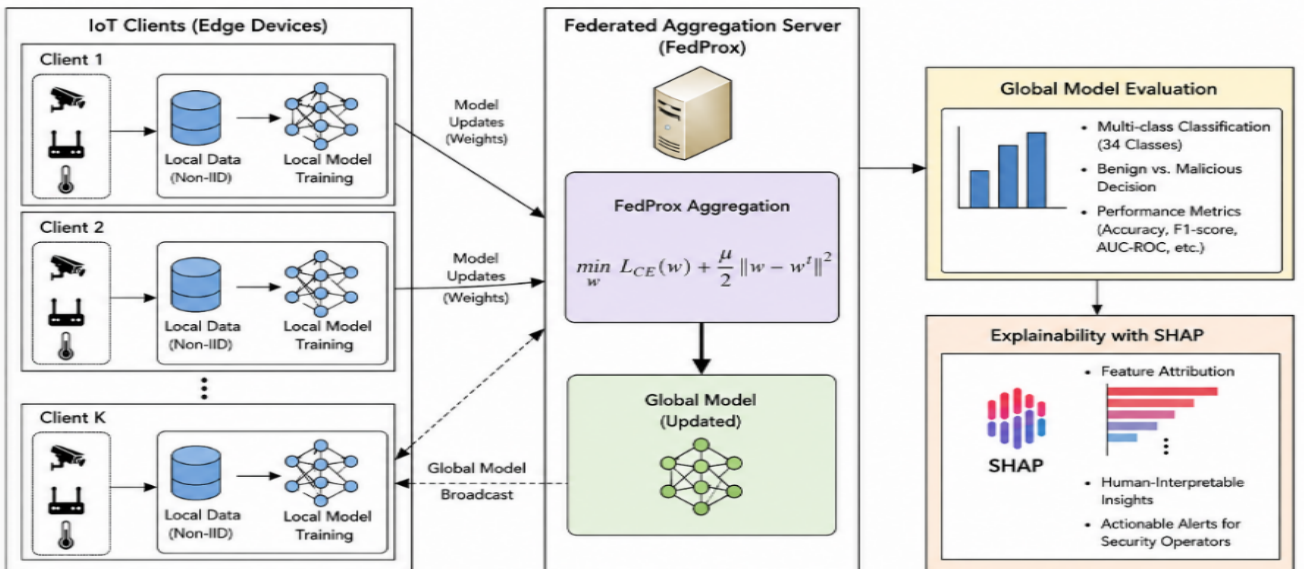


Fig 1. Architecture of the proposed FedShield framework

A. Dataset Curation and Preprocessing

In order to test the proposed framework under realistic IoT attack scenarios, the CIC IoT – 2023 dataset, provided by the Canadian Institute for Cybersecurity (CIC) [10] has been used. The data set includes millions of network traffic cases from 33 attack categories and one benign class and is representative of a wide variety of state-of-the-art cyberattacks, such as DDoS, DoS, Mirai botnet, reconnaissance, spoofing and web-based attacks.

Data preprocessing was done to enhance data quality and ease in efficient model training. Firstly, variance thresholding was applied to remove redundant features to reduce the number of features with low variance (low value). The missing values were filled in with the median and the continuous numerical features were scaled using the Z-score normalization method to guarantee that the features were scaled in the same manner. Finally, the categorical attack labels were converted to the 34 integer encoded classes for multi-class classification using cross-entropy loss function.

B. Simulating Authentic Non-IID Conditions

Traffic patterns are highly variable in real-world IoT settings, as the functionality of the devices, deployment scenarios and user behavior are all different, causing the data that is generated by these clients to be non-Independent and Identically Distributed (non-IID). The preprocessed CIC-IoT-2023 dataset was divided into five virtual clients according to a Dirichlet distribution with concentration parameter $\alpha = 0.5$, to mimic this real-world scenario. In this partitioning approach the distribution of classes is not uniform, some clients have unique set of attack categories, and different proportion of samples.

The resulting data distribution is not uniform across the network which mimics realistic federated IoT deployments and serves as a great challenge to assess the robustness of the proposed FedProx-based federated learning framework to deal with client drift and data heterogeneity.

C. Local Deep Neural Network Architecture

The participating nodes run the same but lightweight Deep Neural Network (DNN) locally on each node. The IDSNet model is designed to be extremely lightweight with only about 90,818 trainable parameters, to consider the stringent requirements on processing and memory capacity when working with edge computing.

The network consists of three fully connected hidden layers (256, 128 and 64 units) and the architectural cascade processes the feature vectors of the network. This is critical to make sure that the model doesn't overfit any local data set since the model is running in a very skewed federated ecosystem. To achieve internal covariate stability, we added Batch Normalization layers right after every linear transformation.

D. Federated Optimization via FedProx

Algorithm 1: FedShield Federated Training using FedProx

Require: K: Number of clients, T: Communication rounds, E: Local epochs, η : Learning rate, μ : Proximal coefficient

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1: Initialize global model weights  $w^0$ 
2: For  $t = 0$  to  $T - 1$  do
3:   Select participating clients  $S_t$ 
4:   For each client  $k \in S_t$  in parallel do
5:      $w_k(t+1) \leftarrow \text{LocalUpdate}(k, w_t)$ 
6:   End For
7:   Aggregate client models
    $w_{t+1} = \sum (n_k / n) w_k(t+1)$ 
8: End For
9: Return global model  $w_T$ 

```

Function LocalUpdate(k, w_t)

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10:  $w \leftarrow w_t$ 
11: For  $e = 1$  to  $E$  do
12:   For each mini-batch  $b \in D_k$  do
13:     Compute Cross-Entropy Loss LCE
14:      $L = \text{LCE} + (\mu/2)\|w - w_t\|^2$ 
15:      $w \leftarrow w - \eta \nabla L$ 
16:   End For
17: End For
18: Return updated local weights  $w$ 
End Function

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Furthermore, we implemented localized models using traditional FedAvg, the severe data skew causes local gradients to point in conflicting directions. Merging these divergent vectors mathematically washes out the global model's intelligence, resulting in catastrophic accuracy loss [8, 17].

Non-IID Data Distribution Across IoT Clients (Dirichlet $\alpha=0.5$)

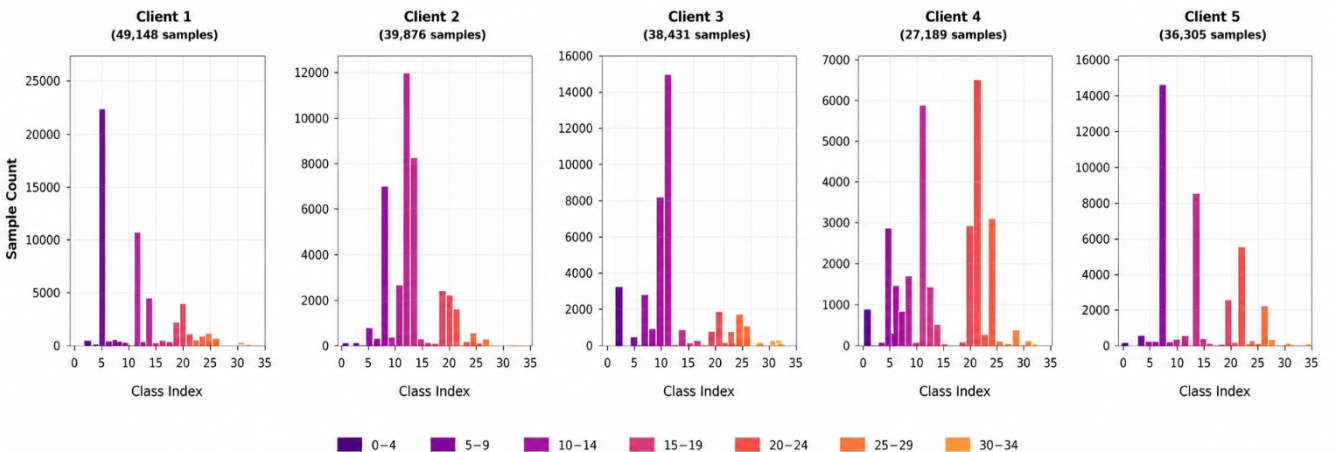


Fig 2. Simulated non-IID data distribution across five IoT clients (Dirichlet $\alpha = 0.5$). The severe, unpredictable class skew mirrors authentic edge network deployments.

FedShield circumvents this phenomenon by deploying the FedProx algorithm. As outlined in Algorithm 1, FedProx restructures the local training objective by introducing a proximal penalty term [18].

$$\min_{w} (LCE(w) + \frac{\mu}{2} \|w - w_t\|^2) \quad (1)$$

In this equation, $LCE(w)$ represents the standard Cross-Entropy loss calculated over the local client's partitioned dataset. The critical control mechanism is the penalty term: μ (which we configured to 0.01) scales the Euclidean distance between the client's currently updating weights (w) and the baseline global weights (w_t) initially broadcasted by the server. With this mathematical tether, a client may wander around and learn from the specific attacks in its locality, but the parameters inside its system cannot deviate from the consensus too far.

E. Explainability via SHAP Integration

Deep learning models have an outstanding detection accuracy but no insight on how they make the decision. To make the model more of an auditable tool, we added SHAPley Additive exPlanations (SHAP) [9] to the model. SHAP is a framework that is inspired by cooperative game theory and allows to measure the exact marginal contribution of each network feature (e.g. packet arrival time, protocol type) to the final output classification. FedShield uses a deep explainer over the aggregated global model to produce clear feature level attributions, which enables human operators to easily confirm any unusual alerts.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

The federated simulation environment was run on Google Colab which was supported by an NVIDIA T4 GPU to enable hardware acceleration. $R = 35$ global communication rounds were run in the global model. All 5 clients were involved in every round, to make sure that the algorithm is tested for its resilience to the Dirichlet skew. The number of local epochs per round was set to 5 on each edge client with a Learning Rate of 0.001 for Adam. All reported metrics were computed on a strictly sequestered 20% of the original data that was held out on a global level to ensure absolute objectivity in reported measures, and avoid data leakage.

A. Convergence Stability and Global Performance

The FedProx algorithm was able to successfully suppress the volatility in the data which is usually seen in non-IID data as can be seen in Fig. 3. The overall test accuracy was steadily increasing, and it passed the initial stabilization stages with ease and reached the highest accuracy of 91.81% at round 34. The computed training time, labeled as total, was about 55.5 minutes, which proved the framework as a computationally efficient one.

TABLE I
Global Model Evaluation Metrics (Peak Performance)

Performance Metric	Achieved Score
Peak Test Accuracy	91.81%
Weighted F1-Score	0.9089
Weighted Precision	0.9131
Weighted Recall	0.8602
Macro AUC-ROC	0.9961
Total Training Time (35 Rounds)	55.5 minutes

The weighted F1-Score of the model followed closely the curve of accuracy, reaching its highest score of 0.9089. Such

a close fit is important in cybersecurity: it is mathematically necessary to prove that the network is not merely learning from the majority class bias of the data set, but is indeed learning complex threat patterns.

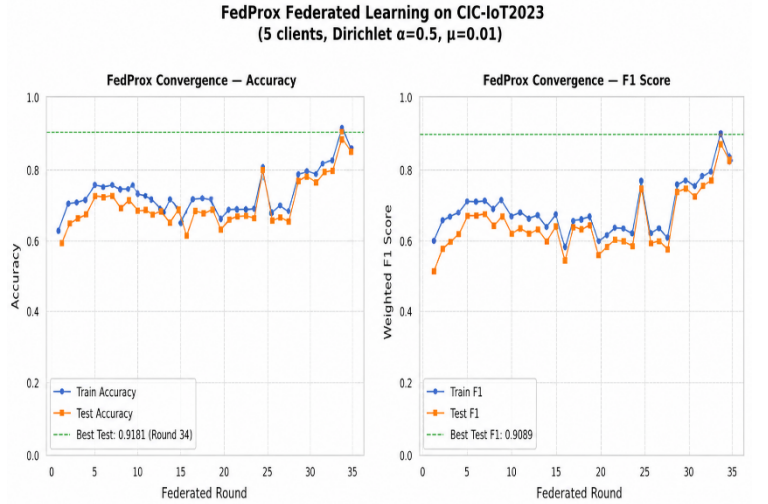


Fig 3. Progress of Global Accuracy and F1-Score on 35 federated rounds. The FedProx proximal penalty distributes the 'chaotic' oscillations that usually occur in the presence of extreme data skews in a clean manner.

B. Diagnostic Performance Across Threat Classes

A high overall accuracy is worthless in the realm of intrusion detection if it does not detect important vectors for intrusion and produces too many false positives. We take a closer look at the global confusion matrix (normalized), and we can see the excellent diagnostic performance in high impact volumetric attacks. The architecture was able to obtain F1-Score of 0.9995 for DDoS-ICMP Flood (more than 7,300 instances) and 0.9998 for DDoS-RSTFIN Flood. Importantly, the system showed good classification rates for the classes of Benign- Traffic with F1-Score: 0.8395. This is the most important feature to avoid the many false positive alerts that network administrators often face that lead to serious "alert fatigue".

But as with all models, one has to look at its shortcomings. One of the main problems with the framework was that it was unable to deal with ultra-minority threat classes. When the number of total exploits of the certain attack type (such as Command- Injection or BrowserHijacking) in our huge testing corpus is below 10, highly sophisticated and rare exploits often yielded an F1-Score of 0.0, as shown in Fig. 4.

This loss of performance is not a failure of the FL architecture, but rather a basic mathematical fact of gradient descent logic – the FL distributed nodes simply didn't have enough data to create differenceable gradient paths for these very rare events.

Even with little data on the extreme minority classes, the overall discriminative power of the model is indisputably strong. This was reflected in the One-vs-Rest Receiver Operating Characteristic (ROC) curves (Fig. 5) which gave an extraordinary Macro AUC of 0.9961. This virtually "perfect" metric demonstrates conclusively that the FedShield architecture was able to accurately identify the "mathematical signatures" of the threats; the framework is able to clearly distinguish attack traffic from benign traffic, across virtually any operational threshold.

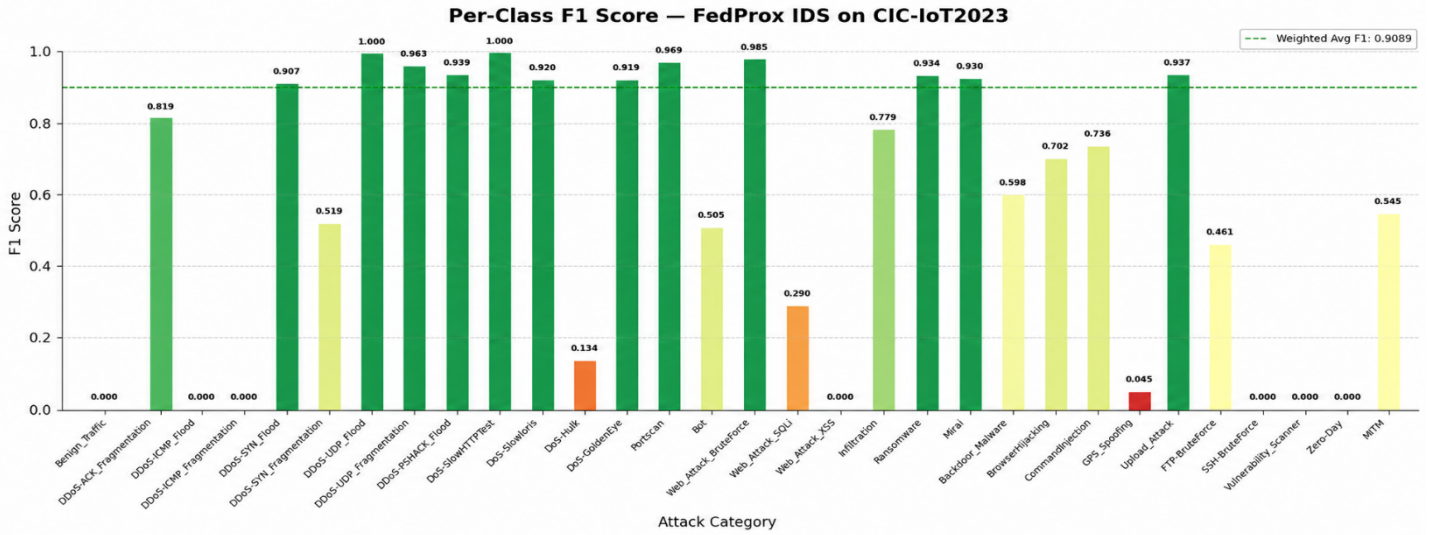


Fig 4. F1-Scores by a given Attack vector. The huge decreases in performance are directly linked to severe data starvation in the ultra-minority classes.

C. Operational Transparency with SHAP

We performed SHAP feature analysis of the final aggregated global model, to turn the FedShield pipeline into a tool that can be used in the industry. Figure 6 very effectively dissects the actual parameters of the network flow that make the neural network classify it as a threat (e.g., Backdoor_Malware). The beeswarm plot also confirms that the network is not making arbitrary decisions based on the random statistical "noise" in the data but instead is relying on the calculated decisions based on the critical network behaviors, including the anomalous PAC intervals, abnormal TCP flag ratios, and abnormal Header Lengths.

This post-hoc transparency is at the finest-grained level that crosses the last hurdle for operations. It enables human security engineers to be able to see the AI's logic, auditable decisions, and deploy AI-driven mitigation responses with confidence, rather than taking a leap of faith.

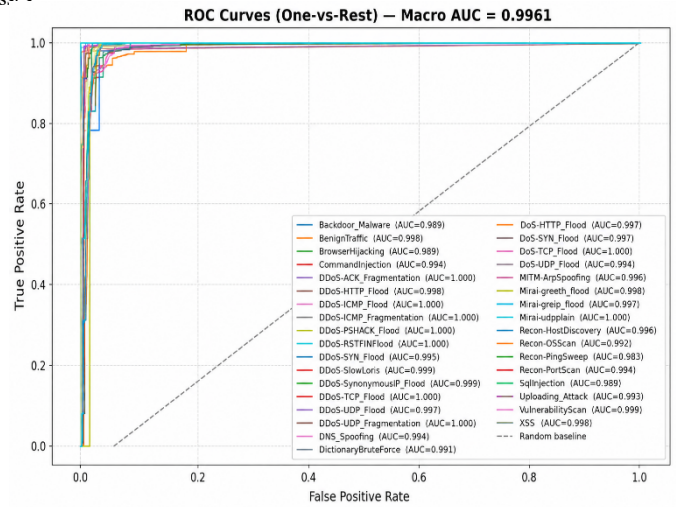


Fig 5. Receiver Operating Characteristic (ROC) curves for one vs rest. The Macro AUC is 0.9961, which is an extraordinary value and which can be considered as a baseline discriminative power of the system.

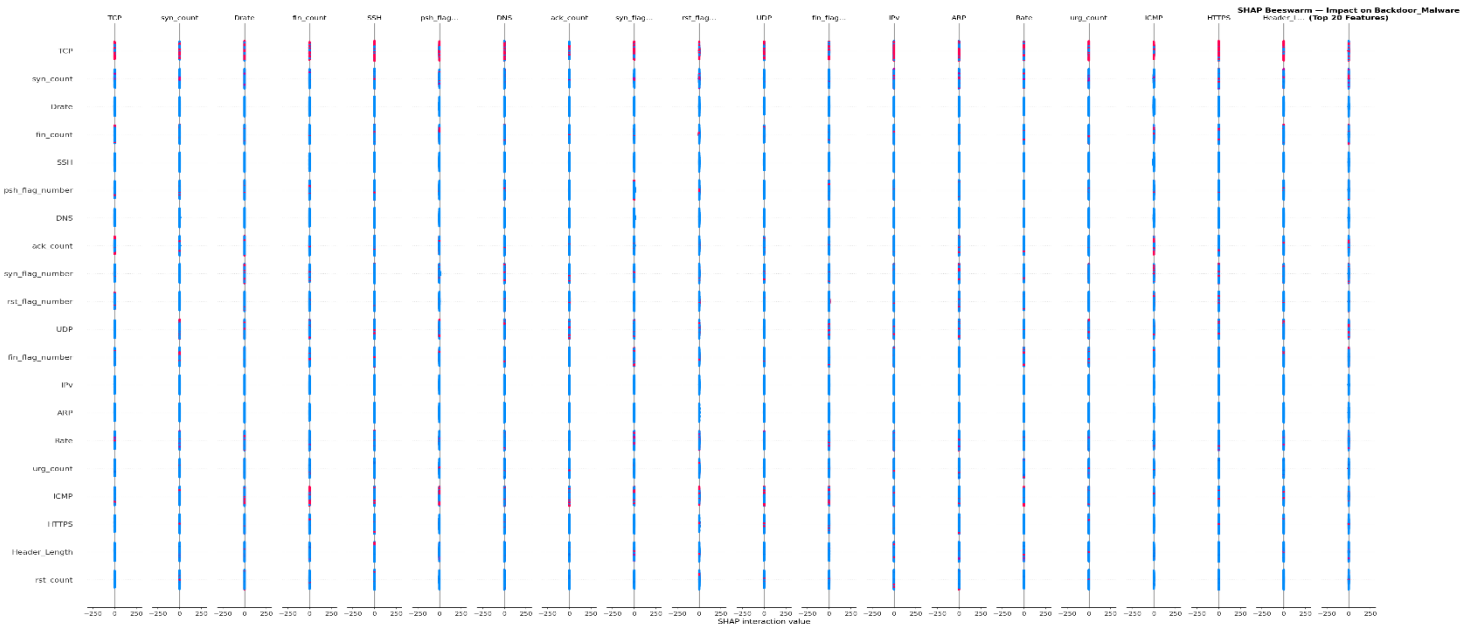


Fig 6. SHAP Beeswarm Plot showing the most important network flow features used to create the model's ID logic for certain malware types.

V. CONCLUSION AND FUTURE WORK

The FedShield framework clearly proves that secure IoT networks can be built, and modern IoT networks can be very decentralised without giving up on data privacy and without consuming a lot of bandwidth. We developed a highly effective TinyML edge architecture that combined with FedProx optimization algorithm, worked well in the harsh non-IID data chaos of real-world edge environments.

Without ever centralizing a single raw data packet, achieving 91.81% test accuracy globally, a weighted F1-Score of 0.9089 and an outstanding Macro AUC of 0.9961 across the 34 different network traffic classes is an immensely positive trade-off between privacy and utility. Moreover, by integrating SHAP directly in the decentralized ecosystem, the Intrusion Detection System becomes a transparent and auditable security tool and is able to be practically deployed in SOC.

Future extensions of this research will focus on exploring few-shot and meta-learning methodologies to dramatically improve the detection of ultra-rare, zero-day attacks at the edge. Additionally, we aim to investigate the integration of cryptographic differential privacy protocols to actively secure the parameter exchange pipeline against sophisticated gradient inference attacks.

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